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Novel Structured Metal Oxide Adsorbent for Efficient Lithium Extraction

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Abstract

Extracting lithium from produced water and brine has gain attention to meet the growing global demand due to the technological advancements included nut not limited with electric vehicles, energy storage systems, and portable electronics. The conventional method for extracting lithium from brine and produced-water involves solar evaporation, where lithium-rich brine is pumped into large evaporation ponds, allowing the water to evaporate and increase the lithium concentration for further processing. However, while this method is commonly used, it suffers from several challenges, including environmental impact, technical complexity, high water consumption, long processing times and operational inefficiencies. Recent advancements have introduced Direct Lithium Extraction (DLE) technologies as a transformative alternative that is providing a more efficient and environmentally friendly alternative to traditional evaporation methods for extracting lithium form brine or produced water. DLE techniques employ various approaches such as adsorption, ion exchange, and solvent extraction to directly extract lithium from complex brines with high ionic concentrations. The shift towards DLE technologies offer faster processing, reduced environmental impact, and greater scalability aligning with sustainability goals and the urgent need for reliable lithium supplies in a rapidly growing market. Here, we focus on development of novel adsorbent system to extract low-concentrated lithium from brine and produced-water. This study presents the development of a novel metal oxide-based adsorbent system specifically engineered to extract lithium from low-concentration brine and produced water. The newly developed adsorbent demonstrates significantly enhanced lithium uptake compared to conventional metal adsorbents particularly under dilute conditions. This improvement is attributed to the material's unique structural and chemical characteristics, including its optimized composition, high surface area, and tailored morphology, which collectively promote more efficient lithium-ion environmentally responsible resource utilization. The findings of this study contribute to the advancement of next-generation lithium extraction technologies and align with global sustainability goals by promoting cleaner, faster, and more reliable methods for securing critical energy materials from unconventional sources such as produced water and low-grade brines.

Introduction

The global technological advancements have boosted lithium to a paramount role in the 21st century, transforming industrial priorities and speeding technological innovation across various sectors. The exponential growth in demand for advanced energy storage systems, more specifically lithium-ion batteries, mainly driven by the electric vehicle revolution, portable electronics and the rapid expansion of renewable energy storage systems, has created an urgent need for more efficient and sustainable lithium extraction technologies (Vega-Muratalla *et al.*, 2024; Busch *et al.*, 2025). Current forecasts indicate that lithium demand is projected to increase ninefold by 2040 (Busch *et al.*, 2025). The scope of lithium applications has expanded significantly beyond the automotive sector to encompass various energy storage applications that are critical for grid stability and renewable energy integration (Vera *et al.*, 2023). Large-scale battery storage systems are being deployed across the United States, Europe, and Asia contributing for making energy storage more consistent and accessible. This diversification of lithium applications has created multiple demand streams that collectively contribute to the exponential growth trajectory observed in recent years (Swain, 2017).

The conventional methods for lithium extraction, specifically hard rock mining and solar evaporation from brine, face significant limitations that limits their ability to meet the projected future demand. Hard rock mining that involves extracting lithium-bearing minerals such as spodumene from pegmatite deposits, requires energy-intensive processing operations including crushing, flotation, and high-temperature conversion processes that typically consume 2-3 times more energy than brine-based extraction methods (Gao *et al.*, 2023). The geographical concentration of these operations further intensifies supply chain liabilities, with Australia, Chile, and China accounting for 90% of global lithium production (Swain, 2017). This concentration creates geopolitical risks and limits the ability to rapidly scale the production in response to demand fluctuations (Busch *et al.*, 2025). In addition, while traditional method that is solar evaporation from brine deposits requires less energy than hard rock mining, it presents its own critical limitations such as restrict its scalability and environmental effects. Traditional solar evaporation method requires large land areas and desired climate conditions. In this method, the brine is stored in large evaporation pond operations typically requiring months to achieve higher lithium concentrations for further processing. Such a long timeline creates inherent inflexibility in responding to market demand changes and limits the technique's applicability to arid regions with high evaporation rates and low precipitation (Swain, 2017). Moreover, there are environmental concerns associated with solar evaporation include significant water consumption, potential impacts on local ecosystems, and the generation of substantial waste brine volumes that may contain elevated concentrations of other minerals and chemicals.

The limitations of traditional methods for Li-extraction are challenging when considering the substantial lithium resources trapped in lower-concentration sources such as produced water from oil and gas operations, geothermal brines, and seawater. These resources offer potentially high lithium reserves that remain mainly undiscovered due to the economic and technical limitations of conventional extraction methods. Produced water from oil and gas operations alone contains estimated lithium concentrations ranging from 10-100 mg/L, and similarly, geothermal brines often contain lithium concentrations of 50-400 mg/L, giving great opportunities for co-production of lithium and other critical minerals that could enhance the economic viability of both operations. The rapid increase in Lithium demand, combined with the intrinsic limitations of conventional extraction methods such as hard rock mining and solar evaporation, has stimulated intensive research for development of more efficient, cost effective and sustainable methods for extraction of lithium from such supplementary resources (Flexer, Baspineiro and Galli, 2018).

Direct Lithium Extraction (DLE) technologies gained attention as an alternative approach for Li-extraction that addresses many of the fundamental limitations associated with traditional extraction methods while offering the potential for more sustainable and efficient resource utilization (Kumar *et al.*, 2019; Farahbakhsh *et al.*, 2024). Different from conventional approaches, which rely on extensive evaporation

or energy-intensive processes, DLE technologies utilize selective separation methods that can operate continuously and meet the market demand. In addition, these DLE systems offer several notable advantages including significantly reduced water consumption, smaller environmental footprints, shortened production timelines from months to hours or days, and the capability to extract lithium from previously non-viable sources including low-concentration brines and produced water streams. The DLE approaches encompasses multiple methods including membrane separation, ion exchange, solvent extraction, and adsorption-based systems (Kumar *et al.*, 2019). Each method offers unique advantages for specific applications. Membrane-based technologies use selective permeability to separate lithium from other dissolved species, while ion exchange systems utilize synthetic resins with preferential affinity for lithium ions. Solvent extraction processes employ organic solvents to selectively extract lithium from aqueous solutions, and adsorption-based systems employ solid materials with selective binding sites for lithium capture. Among these emerging solutions, adsorbent-based systems utilize structural materials for selectively extracting lithium from low-concentration sources, including brines and produced water, even in the presence of competing ions that traditionally complicate extraction processes.

Adsorbent-based DLE systems employ solid materials with selective affinity for capturing lithium ions to achieve separation from complex water compositions such as brine and produced water that is containing multiple competing ions and species. The fundamental mechanism underlying in DLE technologies encompasses the preferential binding of lithium ions to specific active sites on the adsorbent surface or within the adsorbent structure. The adsorption process follows by desorption procedure to recover concentrated lithium solutions suitable for downstream processing. The efficiency of adsorbent-based materials systems depends critically on the material properties of the adsorbent, including selectivity coefficients for lithium over competing ions, adsorption capacity, surface area, porosity, kinetics of uptake and release, and stability under repeated cycling conditions.

The development of efficient and selective adsorbent materials systems for lithium extraction hinders several essential challenges related to the unique properties of lithium ions and the complex compositions of water sources. Lithium is the smallest alkali metal that creates both opportunities and challenges for selective separation, as its small ionic radius enables access to certain binding sites while also making it susceptible to competition from other small ions including sodium and magnesium. The high hydration energy of lithium ions in aqueous solution further complicates selective extraction, as the strong water coordination sphere must be overcome to achieve effective binding to adsorbent sites. These fundamental chemical challenges necessitate careful design of adsorbent materials with binding sites specifically tailored for lithium coordination while minimizing interactions with competing species.

The operational environment of adsorbents for Li-extraction produced water or brine has some materials challenges that effect long-term performance and economic viability. These sources usually contain elevated concentrations of additional elements such as sodium, potassium, magnesium, and calcium ions, along with various anions including chloride, sulfate, and carbonate species that can impact adsorbent efficiency, selectivity and stability. The existence of various competing ions in low-concentration sources cause additional complexity that needs to be addressed through careful adsorbent and process design. Sodium concentrations usually are higher than lithium concentrations that is creating intense competition for adsorbent binding sites and potentially overwhelming lithium-selective mechanisms. Magnesium and calcium ions, while typically present in lower concentrations than sodium, they can create a strong interaction with specific adsorbent materials due to their higher charge density, potentially leading to irreversible binding that reduces adsorbent capacity for lithium over time. Controlling selectivity of various ions requires unique adsorbent materials with exceptional selectivity and process designs that can effectively handle the complex multi-component nature of low-concentration lithium sources. The control over extraction of Li with these competing ion effects requires the design of novel adsorbent materials with

exceptional selectivity and process designs that can effectively handle the complex multi-component nature of low-concentration Li sources.

Metal oxide-based adsorbent materials systems have gained attention as promising materials for selective lithium extraction due to their ability to control over binding of lithium and competing ions based on size, charge, and coordination preferences. Metal-oxide based adsorbents include layered metal oxides, framework structures, and composite materials incorporating metal oxide components, so that they offer the potential for tailored selectivity through tunable pore sizes, surface chemistry, and binding site geometry (Jung and Han, 2013; Swain, 2017; Nicoli *et al.*, 2021; Su, Qian and Qian, 2022; Rajpurohit *et al.*, 2024; Stamenić *et al.*, 2025). The crystalline nature of many metal oxide adsorbents provides structural stability under harsh operating conditions while enabling rational design approaches based on understanding of structure-property relationships.

Metal oxide based adsorbent materials such as aluminum oxide (Al_2O_3) and manganese oxides (MnO_2) have demonstrated exceptional selectivity for lithium extraction through ion-exchange mechanisms that exploit the specific size and coordination requirements of lithium ions (Jung and Han, 2013; Su, Qian and Qian, 2022; Stamenić *et al.*, 2025). These materials usually feature interlayer spaces with dimensions precisely suited for lithium accommodation while excluding larger alkali metal ions through size exclusion effects. The ion-exchange mechanism allows for reversible lithium binding and release through controlled variation of solution pH or ionic strength, enabling effective regeneration and recovery of concentrated lithium solutions. Recent developments in synthesis methods have enabled the preparation of these materials with controlled particle sizes, surface areas, and defect concentrations that optimize both kinetics and capacity for lithium extraction applications (Jung and Han, 2013; Swain, 2017; Flexer, Baspineiro and Galli, 2018; Kumar *et al.*, 2019; Nicoli *et al.*, 2021).

In this study, we aim to address the critical need for improved adsorbent materials for Li-extraction from low-concentration sources through the systematic evaluation of novel structured metal oxide adsorbents designed specifically for high selectivity, and efficiency. The research objectives include development of novel alloy-oxide (AO) adsorbent along with both fundamental understanding of the relationship between adsorbent structure and lithium extraction performance, as well as practical evaluation of material performance under conditions representative of industrial applications.

Experimental

The alloy oxide adsorbent was synthesized using mechanical ball milling as illustrated in Figure 1a. Three different metal oxide components (designated as MO-1, MO-2, and MO-3) were combined in equal mole ratios and subjected to high-energy ball milling. The milling process was conducted using a planetary ball mill that was operating at 1200 rpm for 24 hours to ensure complete mixing and formation of the desired alloy oxide structure. The extended milling duration was selected to advance intimate contact between the different oxide phases and enable the formation of a homogeneous alloy oxide adsorbent material. The resulting alloy oxide powder was characterized for morphological structure, surface area, and pore size distribution prior to adsorption testing. The mechanical alloying process was designed to create a synergistic combination of the individual metal oxide properties while potentially generating new binding sites at the interfaces between different oxide phases.

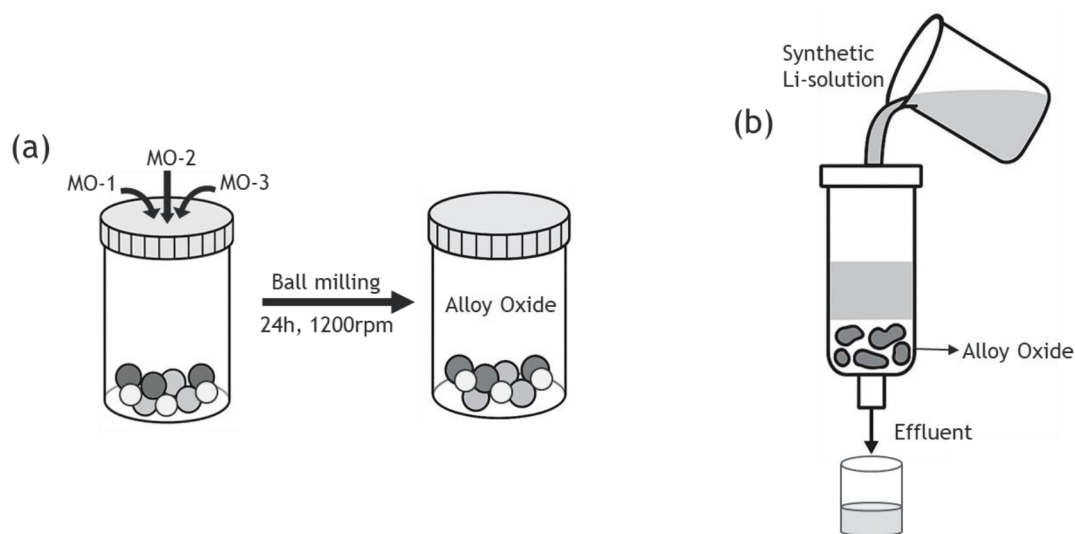


Figure 1—(a) Schematic illustration of alloy oxide adsorbent synthesis by ball milling. Metal oxide precursors (MO-1, MO-2, MO-3) are combined and subjected to high-energy ball milling at 1200 rpm for 24 hours to produce the alloy oxide material. (b) schematic diagram of the fixed-bed column adsorption experiment. Synthetic lithium-containing solution is passed through a column packed with the alloy oxide adsorbent, and the effluent is collected for analysis to evaluate lithium adsorption performance.

The synthesized alloy oxide adsorbent was evaluated for lithium extraction performance using a fixed-bed column setup as shown in Figure 1b. The experimental apparatus consisted of a cylindrical glass column packed with a known mass of the prepared alloy oxide adsorbent. The column was designed to allow controlled flow of synthetic lithium solutions through the adsorbent bed while enabling continuous monitoring of effluent composition.

Synthetic lithium solutions were prepared using LiCl powder. The feed solutions contained lithium ions at 50ppm. The pH of the solutions was systematically adjusted to values of 6, 8, 10, and 12 using diluted hydrochloric acid (HCl) and sodium hydroxide (NaOH) solutions. Each solution was thoroughly mixed and allowed to equilibrate before use to ensure stable pH and ionic strength. All solutions were freshly prepared prior to each experimental run to avoid precipitation or changes in composition. Column adsorption experiments were conducted using a fixed-bed setup, as previously described. For each experimental run, the column packed with the alloy oxide adsorbent was equilibrated to the target temperature (40 °C, 60 °C, and 80 °C). The synthetic lithium solution, adjusted to the desired pH, was then continuously fed through the column at a constant flow rate using a peristaltic pump.

For each temperature and pH condition, effluent samples were collected from the column outlet (Figure 1b). These samples were subsequently analyzed to determine lithium concentrations, allowing for evaluation of adsorption capacity. Between runs at different pH or temperature, the column and adsorbent were thoroughly rinsed and re-equilibrated to avoid cross-contamination and ensure reproducibility.

Results and Discussions

Morphology, surface properties and porosity are critical factors that affect the efficiency and performance of the adsorbents for lithium extraction. To understand morphology and surface properties of the synthesized alloys, we conducted three metal oxide precursor materials labeled as MO-1, MO-2, and MO-3 and an alloy oxide adsorbent synthesized from these precursors. Scanning Electron Microscope (SEM) images at two magnifications (100 μm and 20 μm scales) were analyzed to compare each material's particle morphology, and surface texture. These characteristics are then related to the materials' potential effectiveness as lithium-selective adsorbents. SEM characterization of MO-1, MO-2, MO-3 and their alloy shows clear differences in particle structure and morphology (Fig. 2). MO-1 sample displays a granular morphology with oval

shaped particles (Fig. 2a). The particles are loosely agglomerated, forming a coarse structure. At higher magnification, MO-1 sample's has identical rough and eroded surface. Some fine pores and cracks can be seen on the surface of particles. This fine pore structures and cracks potentially can act as an active site for Li-ion adsorption. MO-2 samples show a noticeably a different structure that is consisting of larger, more defined and merged crystalline particles (Fig. 2b). The particles have faceted, polygonal shapes, forming an ordered morphology that is a characteristic of crystalline aggregates. In addition, MO-2 sample shows much smoother surface compared to MO-1, MO-3 and alloy structure. MO-2 sample particles demonstrate a flat surface with well-defined edges with much less pore structures compared to the MO-1, MO-3 and alloy structure. MO-3 samples display a similar morphological structure to MO-1 with a granular structure of irregular very fine particles (Fig. 2c). At higher magnification, it is clear that very fine particles of irregular and spherical morphology contribute to the interparticle porosity. The mechanical ball mill alloying process transforms the morphologies of the three precursors into a synergistic alloy composite structure where the large MO-2 crystallites, and MO-1 and MO-2 precursors are fractured into smaller the alloy appears as agglomerated clusters that combine granular and crystalline features (Fig. 2d). There are regions of small, irregular particles integrated with larger, facet-like structures that indicates the alloy is a composite of the different oxide phases fully integrated. Overall, the alloy particles show a rough surface that is a result of homogenous integration of precursors that forms the alloy structure. In addition, the alloy's mixed morphology and porosity has a potential for improved lithium extraction performance. Creating a significant pore networks and rough surfaces can enhance the efficiency of Li adsorption. The insights gained from SEM characterization will be used to correlate the adsorbents' physical structure with their lithium adsorption behavior, directing further optimization of the alloy oxide adsorbents. Eventually, controlling the particle morphology, surface properties, and porosity of adsorbents is a fundamental path for developing efficient, selective and durable lithium extraction technologies.

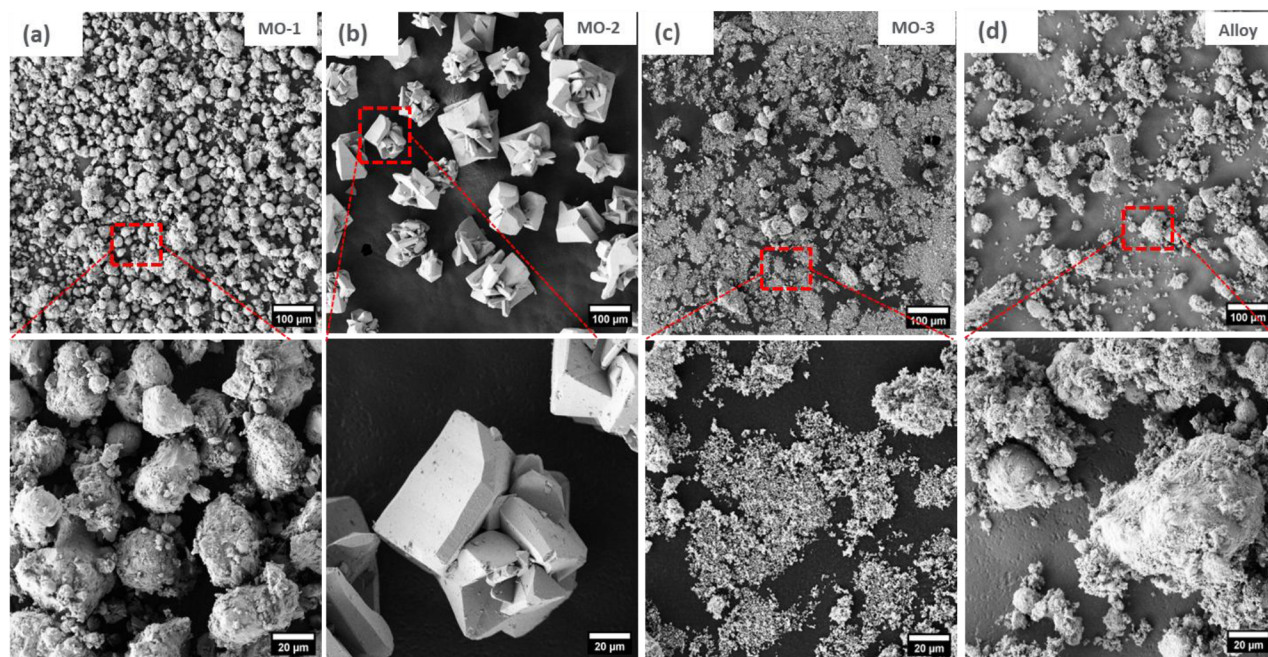


Figure 2—SEM images comparing the morphology of three metal oxide precursors (MO-1, MO-2, MO-3) and the resulting alloy oxide adsorbent. (a) low- and high-magnification of MO-1 samples. (b) low- and high-magnification of MO-2 samples. (c) low- and high-magnification of MO-3 samples and (d) low- and high-magnification of Alloy oxide samples.

Fig. 3 shows the EDX elemental maps of the alloy sample that is synthesized using MO-1, MO-2, and MO-3 precursors. The EDX confirms that alloy particles are uniformly distributed throughout the sample, together with oxygen (green), M1 (red), M2 (blue), and M3 (cyan) metal elements. The grayscale base

image (Ch1) shows the sample morphology and overlaps with the colored maps, confirming that each region of the alloy contains all the key elements homogeneously. There are no distinct areas dominated by a single color, which can be indicated that no elemental segregation is observed. This uniform elemental distribution proves that the precursor components have been successfully integrated into one alloy phase, confirming that the adsorbent material has a consistent composition across its structure.

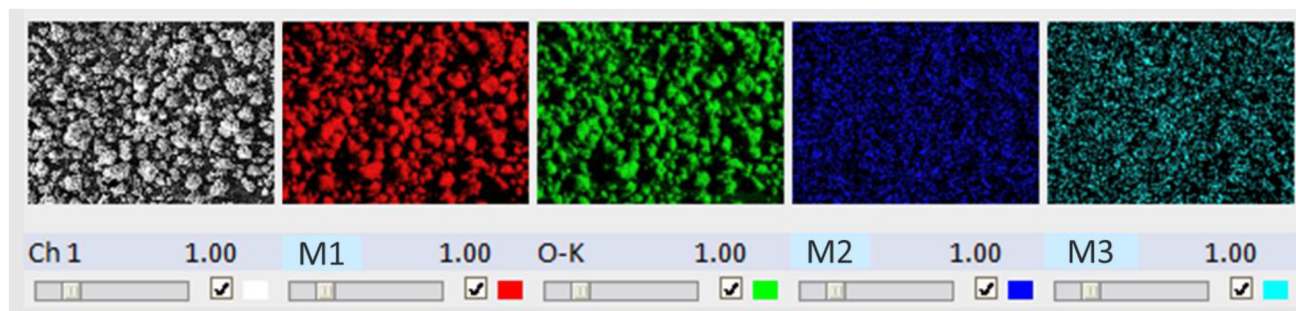


Figure 3—EDX elemental maps of the alloy sample. The grayscale image (Ch1) provides morphological context, while the colored maps indicate elemental signals: M1 (red), O-K (green), M2 (blue), and M3 (cyan). The overlap of red, blue, and cyan signals across the sample demonstrates that all three metal-oxides are homogeneously dispersed through the alloy.

The lithium adsorption capacity of the precursor and alloy oxide (AO) has been conducted at different temperatures and pH for the process optimization (Fig. 4). The alloy oxide adsorbent has shown higher Li-adsorption capacity compared to the individual precursor materials across varying temperatures and pH levels. This enhanced adsorption performance can be correlated to the synergistic effect, where the alloy structure offers capabilities beyond the individual metal oxide structures. Among the precursors, MO1 shows moderate adsorption between 1mg/g to 4mg/g, and MO-2 performs weakly <0.2 mg/g, possible due to its dense crystalline structure limiting lithium access. The alloy oxide demonstrates 20% and 28% improvement compared to the best-performing precursor MO-1, with the effect amplified under alkaline conditions at pH 10 and pH 12 respectively. The alloy oxide has shown the best performance at 80°C and pH 12, where it reaches 7.35 mg/g, which is 28% higher than MO-1. Most probably, the temperature dependency, mainly at pH 12 and 80°C is leading to a structural change that improves the lithium intercalation. The process involves significant activation energy barriers, with elevated temperatures facilitating lithium ion desolvation and diffusion into the alloy. Alkaline pH further enhances performance possibly because of the increased negative surface charge and reduced proton competition that favors the lithium binding on the surface (Ventura, Bhamidi and Hornbostel, no date; Su, Qian and Qian, 2022). These findings support a synergistic mechanism that involves electrostatic interactions with oxide oxygen sites, vacancy/defect-driven Li-intercalation, and possible structural changes under alkaline conditions.

The potential synergistic mechanism here can be state that the lithium adsorption on the alloy oxide primarily is governed by selective intercalation of Li^+ into structural vacancies and strong surface complexation with lattice oxygen sites, facilitated by the engineered structure and composition of the alloy oxide. The alkaline pH favorite supports the proposed mechanism involving lithium interaction with oxide oxygen sites since the higher pH conditions may increase the negative charge density on the surface of oxygen atoms, and enhancing electrostatic attraction for lithium cations (Cipollina *et al.*, 2022). In addition, alkaline conditions has a pottential to reduce competition from protons for binding sites while possibly promoting structural changes in the alloy that increase accessibility to lithium-selective sites (Farahbakhsh *et al.*, 2024). This mechanism potentially enables high selectivity, capacity, and efficient regeneration under varied pH and temperature conditions. Overall, the alloy oxide shows the optimal lithium extraction at 80°C and pH 12, but its strong performance even at milder conditions offers operational flexibility, validating mechanical alloying as a powerful strategy for developing high-efficiency lithium adsorbents.

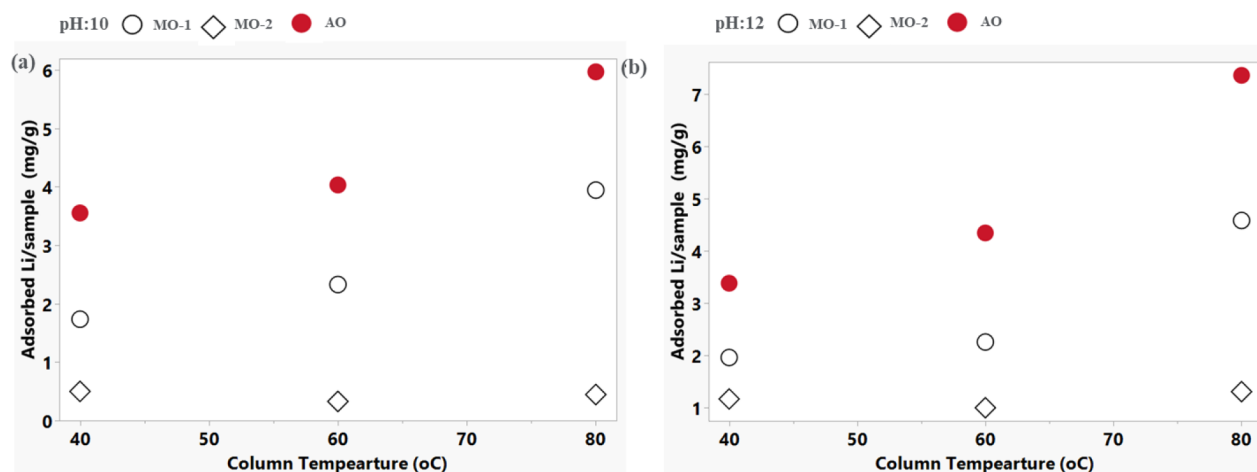


Figure 4—Effect of temperature and pH on lithium adsorption capacity of individual metal oxide precursors (MO-1, and MO-2) and alloy oxide adsorbent (AO). (a) shows results at pH 10, (b) at pH 12. The alloy oxide demonstrates superior performance compared to individual precursors across all tested conditions, with adsorption capacity increasing with both temperature (40–80°C) and pH (10–12).

Conclusion

In conclusion, in this study we demonstrate that the novel alloy oxide-based adsorbent shows a better performance compared to its individual precursor of metal oxides (MO-1, and MO-2,) because of the synergistic combination of structural and chemical advantages. The comprehensive analysis including SEM morphology, EDX elemental mapping, and adsorption performance tests revealed that MO-1 provide higher Li^+ binding sites due to their porous, rough textures, while MO-2 contributes crystalline stability with a limited capacity. The alloy oxide adsorbent achieved an enhanced lithium uptake especially at pH 12 at 80 °C. This improvement can be due to the homogeneous composition of alloy oxide and microstructure, which offer higher active sites, high porosity, and robust phase interfaces for lithium-ion binding. As a result, the alloy oxide adsorbent exhibits better performance in both Li^+ adsorption capacity compared to conventional single-metal oxides. These findings validate the strategy of combining multiple oxides to harness their complementary strengths, high adsorption capacity, and fast ion-exchange kinetics. Overall, the developed alloy adsorbent proves to be a highly effective and durable lithium adsorbent, indicating a promising step forward for direct lithium extraction (DLE) technologies. It doesn't only enhance lithium adsorption efficiency under optimal conditions but also supports sustainable resource utilization by enabling more efficient capture of lithium from dilute brines. The insights gained here provide a clear structure–performance relationship for lithium adsorbents and will guide the design of next-generation materials that further improve lithium extraction efficiency and longevity.

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